

PARTHA PRATIM BHAKAT

Machine Learning Engineer / Software Developer

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PROFESSIONAL SUMMARY

Machine Learning Engineer and Computer Science undergraduate with hands-on experience in **deep learning, Generative AI, agentic LLM systems, and model quality testing**. Proven track record across an AI/ML internship, applied research, and hackathon wins. Seeking a full-time **ML / AI Engineer** role to develop and deploy scalable models.

TECHNICAL SKILLS

Languages: Python, Java, SQL, HTML5, CSS3

AI & Machine Learning: PyTorch, TensorFlow, Keras, CNN, ANN, Deep Learning, NLP, Computer Vision, Hugging Face, OpenCV, LLMs, Generative AI, Agentic AI, Prompt Engineering, LangChain, LangGraph

Data & Web Engineering: NumPy, Pandas, Matplotlib, Seaborn, Next.js, FastAPI, REST APIs, MySQL, MongoDB

Tools & Platforms: Git, GitHub, Docker, VS Code

EXPERIENCE

AI/ML Intern – LLM Testing & Data

Mar 2026 – Apr 2026

Rigzz Technologies

Kolkata, West Bengal

- Executed structured test suites to evaluate LLM output accuracy, formatting integrity, and edge-case behaviors prior to production releases.
- Curated high-quality dataset samples for model benchmarking and validated REST API endpoints for ML microservices to maintain low latency and error handling.

PROJECTS

Deep Agents Conversational AI | *Python, LangChain, LangGraph, DeepAgents, Streamlit*

Aug 2026

- Engineered an agentic chatbot integrating web search, automated planning tools, and persistent virtual memory.
- Implemented context loading, skill routing, and thread-level state memory using LangGraph checkpoints.
- Designed multi-backend architectures with structured Pydantic outputs and automated evaluation workflows.

Real-Time ASL Detection | *Python, PyTorch, OpenCV, TensorFlow, Keras*

May 2026

- Built a real-time American Sign Language recognition system classifying 24 alphabet gestures via webcam feeds.
- Trained a custom CNN on Sign Language MNIST achieving 99.8% training accuracy using batch normalization and dropout.
- Integrated OpenCV for computer vision tasks including background subtraction and real-time hand contour extraction.

EDUCATION

Vellore Institute of Technology (VIT)

Aug 2023 – May 2027

B.Tech in Computer Science and Engineering – CGPA: 7.74

Bhopal, India

PUBLICATIONS & PATENTS

- Patent:** Self-Recharging System for an Unmanned Aerial Platform
- Publication:** Entertainment Multimedia Compression Using YOLOv8 and GAN – *Under review, Springer, 2026*

LEADERSHIP & ACHIEVEMENTS

- 2nd Place Winner** – Innovation Hackathon for Mahakumbh MP 2028 Preparedness (Kumbh-Cortex Omni-Flow), *Jun 2026*
- Operations Lead** – The Bengali Club, VIT Bhopal (Managed event workflows and logistics), *Mar 2024 – Mar 2025*
- Core Member, Marketing** – Anterix Club, VIT Bhopal (Executed targeted campaigns to boost outreach), *May 2025 – May 2026*
- Certified: Complete Data Science, ML, DL, & NLP Bootcamp** – *Udemy, 2025*

CORE COMPETENCIES

Teamwork & Collaboration · Problem-Solving · Adaptability · Time Management · Leadership & Mentoring